MUHAMMAD AQEEL

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SUMMARY

I am a computer vision researcher who adapts foundation models to problems where labeled data is scarce. My work covers vision-language adapters for zero-shot detection (CLIP, DINOv3), generalist models that transfer to unseen domains, diffusion-based image synthesis, and video summarization with vision-language models, including hallucination auditing. I have first-author papers at NeurIPS 2026, ICCV 2025 and ICIP 2026. At Verona I lead a research line on generalist anomaly detection, have supervised five master's students, and collaborate with industrial partners on visual inspection. Before moving into research, I spent two years in software development and quality assurance.

EXPERIENCE

Postdoctoral Research Fellow Mar. 2026–Present
University of Verona *Verona, Italy*

- Lead the generalist anomaly detection line: DART (NeurIPS 2026) separates domain and defect information in the residuals of frozen DINOv2, CLIP and WideResNet backbones, as a plug-in for three state-of-the-art detectors; +13.3 AUROC when transferring from industrial images to brain MRI.
- Video understanding with vision-language models (with M. Nazir): summarization of instructional videos (ICPR 2026) and a reproducibility study of grounded hallucination auditing in video summaries (ICPR 2026 workshop).
- Researcher on spindle5.0, an industrial anomaly detection project (Cari Verona).
- Supervise five master's students, including a thesis extending CoMet to MVTec AD 2.

Ph.D. Researcher (PNRR Doctoral Fellowship) 2022–2026
University of Verona *Verona, Italy*

- Zero-shot image understanding with industrial partner Qualyco: AVA-DINO (ICIP 2026), dual vision-language adapters on frozen DINOv3 and CLIP; 93.5% image AUROC on MVTec AD with 28M trainable parameters.
- Learning from uncurated data with Qualyco: CoMet (ICCV 2025) weights each training image by confidence; recall rose from 76.4% to 93.4%, and results held with up to 10% defective images in the training set.
- Fine-tuned LLaVA-style vision-language models to reason step by step about defects.
- Synthetic training data: conditioned pre-trained latent diffusion models (image, mask and text prompt) to generate defects for detector training (ExDD, RoadFusion; ICIAP 2025), including road inspection with the Comune di Ferrara.

Visiting Researcher Jun.–Nov. 2025
University of Ljubljana *Ljubljana, Slovenia*

- Zero-shot anomaly detection with Prof. Danijel Skočaj's group: CoZAD, which combines contrastive learning with confident meta-learning (ICCV 2025 VISION workshop); funded by a mobility research grant.

Teaching Assistant, Computer Vision and Machine Learning 2023–2025
University of Verona *Verona, Italy*

Research Assistant, Natural Language Processing (Sentiment Analysis) 2021–2022
Huazhong University of Science & Technology *Wuhan, China*

Software Quality Assurance Engineer Sep. 2018–Aug. 2019
UIIT, PMAS Arid Agriculture University *Rawalpindi, Pakistan*

- Wrote test cases from requirements, ran functional and regression tests, and verified fixes with the developers.

Software Developer Sep. 2017–Aug. 2018
eConceptions (Pvt) Ltd *Islamabad, Pakistan*

- Built and maintained product features and fixed bugs in a small development team.

EDUCATION

Ph.D. in Computer Science 2022–2026
University of Verona · Supervisor: Prof. Francesco Setti *Verona, Italy*

M.Sc. in Computer Science 2019–2021
Beijing University of Posts & Telecommunications · Supervisor: Prof. Xiaohong Liu *Beijing, China*

B.Sc. in Computer Science 2013–2017
PMAS Arid Agriculture University *Rawalpindi, Pakistan*

PUBLICATIONS

Conference Proceedings

[18] DART: Domain-Agnostic Residual Transfer for Generalist Anomaly Detection — NeurIPS–2026
***M. Aqeel**, M. Nazir, M. Cristani & F. Setti*
Advances in Neural Information Processing Systems (Accepted)

[17] Anomaly-Aware Vision-Language Adapters for Zero-Shot Anomaly Detection — ICIP–2026
***M. Aqeel**, M. Nazir, U. Khan, M. Cristani & F. Setti*
IEEE International Conference on Image Processing

[16] Multimodal Abstractive Summarization of Instructional Videos with Vision-Language Models — ICPR–2026
*M. Nazir, R. Zhang, **M. Aqeel** & F. Setti*
International Conference on Pattern Recognition

[15] LiZAD: A Lightweight Zero-Shot Anomaly Detection Framework for Industrial Manufacturing — COINS–2026
*U. Khan, L. Capogrosso, **M. Aqeel**, F. Setti, M. Magno & M. Cristani*
IEEE International Conference on Omni-layer Intelligent Systems

[14] Auditing the Auditor: A Reproducibility Study of Grounded Hallucination Auditing in Video Summarization — ICPRW–2026
*M. Nazir, **M. Aqeel**, M. Cristani & F. Setti*
International Conference on Pattern Recognition, RRPR Workshop

[13] Towards Real Unsupervised Anomaly Detection via Confident Meta-Learning — ICCV–2025
***M. Aqeel**, S. Sharifi, M. Cristani & F. Setti*
Proceedings of the IEEE/CVF International Conference on Computer Vision

[12] Robust Anomaly Detection in Industrial Environments via Meta-Learning — ICCVW–2025
***M. Aqeel**, S. Sharifi, M. Cristani & F. Setti*
Proceedings of the IEEE/CVF International Conference on Computer Vision, VISION Workshop

[11] A Contrastive Learning-Guided Confident Meta-Learning for Zero-Shot Anomaly Detection — ICCVW–2025
***M. Aqeel**, D. Skočaj, M. Cristani & F. Setti*
Proceedings of the IEEE/CVF International Conference on Computer Vision, VISION Workshop

[10] ExDD: Explicit Dual Distribution Learning for Surface Defect Detection via Diffusion Synthesis — ICIAP–2025
***M. Aqeel**, F. Leonardi & F. Setti*
International Conference on Image Analysis and Processing, Springer

[9] RoadFusion: Latent Diffusion Model for Pavement Defect Detection — ICIAP–2025
***M. Aqeel**, K. D. Bellete & F. Setti*
International Conference on Image Analysis and Processing, Springer

[8] Latent Space Synergy: Text-Guided Data Augmentation for Direct Diffusion Biomedical Segmentation — ICIAPW–2025
***M. Aqeel**, M. Nazir, Z. Ruan & F. Setti*
International Conference on Image Analysis and Processing, Workshops, Springer

[7] Self-Supervised Iterative Refinement for Anomaly Detection in Industrial Quality Control — VISAPP–2025
***M. Aqeel**, S. Sharifi, M. Cristani & F. Setti*
Proc. 20th Int. Joint Conf. on Computer Vision, Imaging and Computer Graphics Theory and Applications, SCITEPRESS, pp. 173–183

[6] Diffusion-Based Data Augmentation for Medical Image Segmentation — ICCVW–2025
*M. Nazir, **M. Aqeel** & F. Setti*
Proceedings of the IEEE/CVF International Conference on Computer Vision, CVAMD Workshop

[5] Meta Learning-Driven Iterative Refinement for Robust Anomaly Detection in Industrial Inspection — ECCVW–2024
***M. Aqeel**, S. Sharifi, M. Cristani & F. Setti*
European Conference on Computer Vision, Workshops, Springer, pp. 445–460

[4] Self-supervised Learning for Robust Surface Defect Detection — DeLTA–2024
***M. Aqeel**, S. Sharifi, M. Cristani & F. Setti*
International Conference on Deep Learning Theory and Applications, Springer, pp. 164–177

[3] Diffusion-Based Image Generation for In-Distribution Data Augmentation in Surface Defect Detection — VISAPP–2024
*L. Capogrosso, F. Girella, F. Taioli, M. D. Chiara, **M. Aqeel**, F. Fummi, F. Setti & M. Cristani*
Proc. 19th Int. Joint Conf. on Computer Vision, Imaging and Computer Graphics Theory and Applications, SCITEPRESS, pp. 409–416

Journal Articles

[2] Knowledge Graph Attention Mechanism for Factually Consistent Multimodal Summarization NCA–2026
M. Nazir & ***M. Aqeel***
Neural Computing and Applications (accepted for publication)

[1] Semantic Parsing for Aspect-Based Sentiment Analysis IEEE Access–2025
M. Aqeel *& F. Setti*
IEEE Access, Vol. 13, pp. 127322–127334

ACADEMIC SERVICE & TALKS

Reviewer: ICCV, ECCV, AAAI, IJCAI-ECAI, BMVC, VISAPP; Pattern Recognition, IEEE Trans. Industrial Informatics, IEEE Trans. Multimedia, Neural Computing and Applications, Applied Soft Computing.
Talks: orals at ICCV 2025 (VISION workshop, ×2), ICIAP 2025 (×2), VISAPP 2025, ECCV 2024 (VISION workshop) and DeLTA 2024; invited talk at ICIAP 2025 (T4S workshop).

FUNDING & GRANTS

PNRR Doctoral Fellowship, University of Verona 2022–2026
Mobility Research Grant, University of Ljubljana 2025
Chinese Government Scholarship (CSC), M.Sc., Beijing University of Posts & Telecommunications 2019–2021

TECHNICAL SKILLS

Programming	Python, C, C++, Java, MATLAB
Frameworks	PyTorch, Hugging Face; TensorFlow and Keras (earlier projects)
Models	CLIP, DINOv2 and DINOv3, ViT, LLaVA (fine-tuning), latent diffusion (conditioning and inpainting), BART
Tools	Git, Docker, Linux, LaTeX, VS Code

LANGUAGES

Urdu / Punjabi (native) · English (proficient, C2) · Italian (basic, A2) · Chinese (basic, A2)